Lib-CSE

Reg. No. [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] **[18ESCIC31]**

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2021

DISCRETE MATHEMATICS AND GRAPH THEORY

Time: 3 Hours **Maximum Marks: 70**

Instructions: **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.

ii. Missing data may be suitably assumed.

1. a. Define the following terms with an example for each **04**

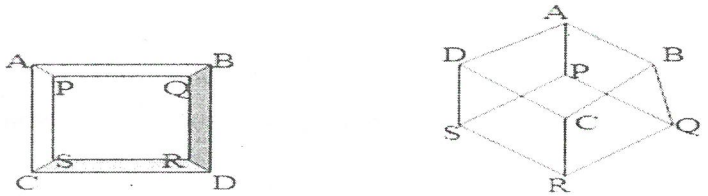
i. Walk

ii. Trail

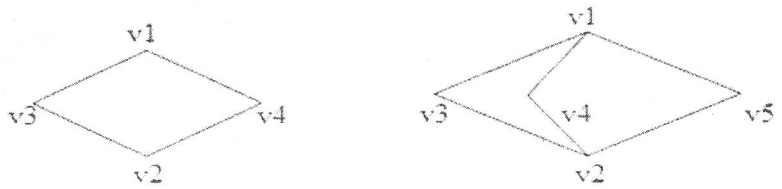
iii. Circuit

iv. Cycle.

b. Define Isomorphism with example. Verify that the given graphs are isomorphic **05**



c. Define the Planar Graph and Non planar graph. Show that the bipartite graph K2,2 and K2,3 are planar graph **05**



2. a. Prove that a graph of order n≥2 consisting a single cycle in 2 chromatic if n is even, and 3 chromatic if n is odd. **04**

b. Find the number of permutations of the letters of the word MASSASAUGA. In how many of these, all four A's are together? How many of them begin with S? **05**

c. The independent term of x is 80000 in the expansion of $\{3x+ (b/x)\}^6$, where b is a positive constant. What the value of b? **05**

3. a. A bag contains 4 white and 2 black balls. Another bag contains 3 white and 5 black balls. If a ball is drawn from each, find the probability that **05**

i. Both are white

ii. Both are black

iii. One is white and another is black?

iv.

b. In a school 25% of the students failed in first language, 15% of the students failed in second language and 10% of the students failed in both. If a student is selected at random find the probability that **05**

i. He failed in first language if he had failed in second language.

ii. He failed in second language if he had failed in first language.

iii. He failed in either of the languages?

c. State and prove addition theorem of probability for mutually and not mutually exclusive events? **04**

OR

4. a. Let A and B be finite sets with $n(A)=m$ and $n(B)=n$. **04**

i. Find how many functions are possible from A to B?

ii. If $A=\{1,2,3\}$ and $B=\{w,x,y,z\}$, find the number of functions from A to B and B to A.

iii. If there are 2187 functions from A to B and $n(B)=3$ then what is $n(A)$?

b. There are six programmers who can assist eight executives. In how many ways can the executives be assisted so that each programmer assists at least one executive? **05**

c. Let A=R be the set of real numbers and $B=\{x/x \text{ is real and } x\geq 0\}$. A function $f:A\rightarrow B$ is defined by $f(a)=a^2$. Verify whether f is one-one and onto or not? **05**

5. a. Among 100 students , 32 study mathematics , 20 study physics , 45 study biology , 15 study mathematics and biology , 7 study mathematics and physics , 10 study biology and physics and 30 do not study any of the subjects. **05**

i. Find the number of students studying all three subjects?

ii. Find the number of students studying exactly one of three subjects?

b. Define derangements. For the positive integers 1,2,3.....*n*, there are 11660 derangements where 1,2,3,4 and 5 appear in the first five positions. What is the value of *n*? (given: d_6=265) **05**

c. Find the Rook polynomial for the following boards for the non-shaded squares **04**

2	3	4
1		5

OR

6. a. Out of 30 students in a hostel, 15 students study Mathematics,8 study Physics, 6 study Chemistry. It is known that 3 students study all these subjects. Show that seven or more students study none of these subjects. **05**

b. How many integers between 1 and 300 (inclusive) are **05**

i. Divisible by at least one of 5, 6, 8?

ii. Divisible by none of 5, 6, 8?

c. Find the number of derangements of 1,2,3,4. **04**

7. a. Find the sequences generated by the function of (i) $3x^3 + e^{2x}$ (ii) $(3+x)^3$. **07**

b. Find the convolution of the sequences $0,1,2,3,4.....$ and $0^2,1^2,2^2,3^2,4^2.....$ Also find the generating function for their convolution. **07**

OR

8. a. Solve the recurrence relation $a_{n+2} - 4a_{n+1} + 3a_n = -200$ for $n \geq 0$ and $a_0 = 3000$, $a_1 = 3300$, by the method of generating functions. **07**

b. Solve the recurrence relation $F_{n+2} = F_{n+1} + F_n$ for $n \geq 0$, given that $F_0 = 0$ and $F_1 = 1$. **07**